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Systems for extracting solute from a source material

Abstract

A system is provided that includes a solvent source; a canister adapted to contain plant material and positioned downstream of the solvent source; an extract container positioned for receiving and collecting at least a portion of an extract solution from at least the canister, the extract container further positioned to separate at least a portion of the solvent from at least the portion of the extract solution, thereby providing a recycled solvent; a solvent collection line for directing the recycled solvent; and a coolant line positioned for directing coolant that is thermally coupled to at least a portion of the solvent collection line for cooling the recycled solvent that is flowing through the solvent collection line.

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References Cited

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2011/0133120	12/2010	McGhee	422/284	C11B 1/10

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of and claims priority under 35 U.S.C. § 120 of U.S. patent application Ser. No. 16/726,057 filed on Dec. 23, 2019 and titled SYSTEMS FOR EXTRACTING SOLUTE FROM A SOURCE MATERIAL, which in turn is a continuation application of and claims priority under 35 U.S.C. § 120 of U.S. patent application Ser. No. 15/910,536, now U.S. Pat. No. 10,595,555, issued Mar. 24, 2020 filed on Mar. 2, 2018 and titled METHODS FOR CREATING CONCENTRATED PLANT MATERIAL SOLUTIONS, which in turn is a continuation-in-part application of and claims priority under 35 U.S.C. § 120 of U.S. patent application Ser. No. 14/157,418, now U.S. Pat. No. 9,926,513, issued Mar. 27, 2018 filed on Jan. 16, 2014 and titled METHODS FOR CREATING CONCENTRATED PLANT MATERIAL SOLUTIONS, which in turn is a continuation-in-part application of and claims priority under 35 U.S.C. § 120 of U.S. patent application Ser. No. 14/070,972, now U.S. Pat. No. 9,145,532, issued Sep. 29, 2015 filed on Nov. 4, 2013 and titled METHODS FOR EXTRACTING SOLUTE FROM A SOURCE MATERIAL,

which in turn is a continuation-in-part application of and claims priority under 35 U.S.C. § 120 of U.S. patent application Ser. No. 14/070,942, now U.S. Pat. No. 9,144,751, issued Sep. 29, 2015 filed on Nov. 4, 2013 and titled SYSTEMS FOR EXTRACTING SOLUTE FROM A SOURCE MATERIAL. The contents of these applications are incorporated herein by reference.

BACKGROUND

(1) The present disclosure relates generally to systems for extracting solute from source materials. In particular, systems configured to extract essential oils from solid materials are described.

(2) Known extraction systems are not entirely satisfactory for the range of applications in which they are employed. For example, many conventional systems are unable to simultaneously extract solute from materials stored in a plurality of distinct containers. Rather, many existing systems are configured to extract solute from a single container of source material.

(3) Extracting solute from a single container produces a bottleneck, requiring the user to wait for extraction to complete before the user can perform any other task associated with the extraction process. As a result, conventional systems require a wasteful, inefficient extraction methodology. Accordingly, there exists the need for a system that provides parallelism to overcome such bottlenecks.

(4) Further, many conventional extraction systems do not allow extraction to be performed in a single, closed loop process that reclaims solvent and re-introduces the reclaimed solvent in subsequent cycles of the system. While some conventional systems allow users to manually reclaim and reuse solvent, this process is time consuming and results in an inefficient reclaim rate.

(5) Further, many systems include no means for reclaiming previously used solvent. Because users are unable to reclaim and reuse solvent, such systems are wasteful. Further, many solvents include odorants and other impurities that may end up in extracted materials. Because the reclamation process may simultaneously purify previously used solvent, systems lacking reclamation functionality may produce a lower quality end product. Accordingly, there exists the need for a system that defines a closed loop with an at least partially automated means for reclaiming and reintroducing solvent used in previous iterations of extracting solute from a source material.

(6) Additionally, or alternatively, there exists a need for cooling reclaimed solvent to a liquid state prior to collecting the solvent. Liquid solvent may be more efficiently stored than solvent that is presently a gas. As a result, cooling reclaimed solvent to a liquid state prior to collecting it allows users to more efficiently store reclaimed solvent. Accordingly, there exists a need for extraction systems that cool reclaimed solvent prior to collecting it.

(7) Thus, there exists a need for extraction systems that improve upon and advance the design of known systems. Examples of new and useful extraction systems relevant to the needs existing in the field are discussed below.

SUMMARY

(8) The present disclosure is directed to systems for extracting solute from a source material. A system may include: a first fluid pathway comprising a solvent source positioned to receive and provide solvent and at least one solvent conditioner in fluid communication with the solvent source and positioned to receive liquid solvent, condition received liquid solvent by maintaining or establishing at least one of a pressure and temperature of the received liquid solvent, and provide conditioned liquid solvent at the at least one pressure and temperature.

(9) At least one canister is adapted to contain plant material, the at least one canister is positioned downstream of the first fluid pathway for receiving the conditioned liquid solvent, thereby producing an extract solution of the conditioned liquid solvent and the essential oil extracted from the plant material. A canister interface may be configured to detachably support the at least one canister and at least one extract container may be positioned to be in selective fluid communication

with the at least one canister for receiving and collecting the extract solution from the at least one canister.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a diagrammatic view of a first example of a system for extracting solute from a source material.
- (2) FIG. 2 is a perspective cutaway view of an example of a detachable canister included in the system shown in FIG. 1.
- (3) FIG. 3 is a perspective view an extract container included in the system shown in FIG. 1.
- (4) FIG. 4 is a cross-sectional view of the extract container shown in FIG. 3 taken along the line 4-4.
- (5) FIG. 5 is a perspective view of a condenser column shown included in the system shown in FIG. 1, the condenser column shown with a front panel removed to show interior details.
- (6) FIG. 6 is a diagrammatic view of a second example of a system for extracting solute from a source material.
- (7) FIG. 7 is a flow diagram of an example of a method for extracting solute from a source material.

DETAILED DESCRIPTION

- (8) The disclosed systems will become better understood through review of the following detailed description in conjunction with the figures. The detailed description and figures provide merely examples of the various inventions described herein. Those skilled in the art will understand that the disclosed examples may be varied, modified, and altered without departing from the scope of the inventions described herein. Many variations are contemplated for different applications and design considerations; however, for the sake of brevity, each and every contemplated variation is not individually described in the following detailed description.
- (9) Throughout the following detailed description, examples of various systems are provided. Related features in the examples may be identical, similar, or dissimilar in different examples. For the sake of brevity, related features will not be redundantly explained in each example. Instead, the use of related feature names will cue the reader that the feature with a related feature name may be similar to the related feature in an example explained previously. Features specific to a given example will be described in that particular example. The reader should understand that a given feature need not be the same or similar to the specific portrayal of a related feature in any given figure or example.
- (10) With reference to FIGS. 1-5, a first example of a system for extracting solute from a source material, system **100**, will now be described. As FIG. 1 shows, system **100** includes a solvent source container **120**, a solvent compressor **130**, a detachable canister system **140**, an extract container **170**, a first pump **101**, a second pump **102**, a condensing system **105**, and a solvent collection container **115**. System **100** additionally includes various valves and fluid lines (defining pipes) that control the flow of fluids through system **100** during operation.
- (11) System **100** may be particularly adapted for using butane to extract essential oils from plant material. For example, FIG. 2 depicts system **100** using butane to extract essential oils from lavender plants. FIG. 2 illustrates lavender **91** contained within first detachable canister **150**, being exposed to a solvent, defining liquid butane **92**, within a canister of system **100**.
- (12) As FIG. 2 shows, first detachable canister **150** is configured to store lavender **91** and liquid butane **92** to extract the essential oils from the source material in an extract solution, containing butane and lavender essential oil. After a predetermined period of time selected to effectively extracting essential oil from lavender plant materials, first detachable canister **150** is configured to

output the contained extract solution.

(13) After the predetermined period of time, the extract solution is directed to an extract container. The extract container is configured to heat the contained extract solution above the boiling point of the solvent to separate substantially purified post-extraction solvent from the extract solution. The evaporated post-extraction solvent is then stored to reclaim it for later use. After removing the post-extraction solvent from the extract container, the residual material in the extract container defines a distilled, high-purity essential oil of the source material.

(14) After the solvent has been used to extract solute from the solvent, system **100** is configured to reclaim the used solvent for later use. As FIG. **1** illustrates, extract container **170** is connected in fluid communication with solvent collection container **115**. Extract container **170** is configured to separate the solvent from the extracted solute, allowing system **100** to direct and collect the used solvent in solvent collection container **115**. System **100** includes several features configured to increase the reclaim rate of post-extraction solvent, allowing system **100** to use solvent more efficiently than many conventional extraction systems.

(15) As FIG. **1** shows, solvent source container **120** is connected in fluid communication with solvent compressor **130** and in fluid communication with solvent collection container **115**. As FIG. **1** illustrates, solvent source container **120** includes a source container input **121** and a source container output **122**. Source container input **121** is configured to fluidly receive solvent communicated from solvent collection container **115**. For example, first pump **101** and second pump **102** may pump solvent contained in solvent collection container **115** as system **100** proceeds through an extraction cycle. Source container input **121** is additionally configured to restrict the passage of fluid back into solvent collection container **115**.

(16) Source container output **122** is configured to direct solvent contained in solvent source container **120** to solvent compressor **130** via a solvent source line **197**. By directing fluid to solvent compressor **130**, solvent source container **120** introduces the solvent in the current cycle of system **100**'s extraction process. In some examples, solvent source line **197** may include an internal filter. The internal filter may be used to remove impurities in solvent prior to introducing the solvent to detachable canister system **140**.

(17) Because solvent source container **120** is configured to fluidly receive solvent from solvent collection container **115**, solvent source container **120** may be refilled with post-extraction solvent collected by solvent collection container **115** during previous extraction cycles performed by system **100**.

(18) By directly reintroducing post-extraction solvent to solvent source container **120**, system **100** is able to reclaim post-extraction solvent at a high rate. Further, the reclaimed solvent may be of a higher purity than fresh, commercially sourced butane. Butane often ships with an odorant, such as mercaptan or thiophane. When using a solvent containing such an odorant, the extracted essential oil may include portions of the odorant. This results in a less desirable end product.

(19) In some examples, solvent source line **197** may include a solvent filter within its fluid-transmissive interior, thereby passing solvent through the filter as it passes from solvent source container **120** to solvent compressor **130**. In some examples, the solvent filter may define a 13-X molecular sieve configured for membrane filtration of the solvent as it passes from solvent source container **120** to solvent compressor **130**.

(20) Post-extraction solvent that has been processed and reclaimed by system **100** may have decreased levels of odorant compared to commercially available odorant-containing solvents. Accordingly, using reclaimed solvent may result in a purer, more desirable end product. In some cases, users may run a solvent purification cycle prior to extraction to remove such impurities. Such a solvent purification cycle may include processing and reclaiming commercially purchased butane through system **100** one time prior to extraction.

(21) As FIG. **1** shows, solvent compressor **130** is in fluid communication with solvent source container **120**. As FIG. **1** additionally illustrates, solvent compressor **130** is in fluid communication

with first pump **101** and second pump **102**, assuming appropriate valves are open. Solvent compressor **130** is configured to receive solvent from solvent source container **120**.

(22) Solvent compressor **130** is configured to compress, or “charge,” the received solvent. In some examples, the compressor may be electrically powered, such as by plugging into an electrical outlet **89**. In other examples, solvent compressor **130** may pressurize solvent using backflow pressure produced by first pump **101** and second pump **102**.

(23) In some examples, it is desirable to use a high temperature, liquid solvent for extraction. Solvent compressor **130** may be used to compress solvent to an extraction pressure, the extraction pressure selected to maintain solvent in a liquid state even when exposed to an elevated extraction temperature. After pressurizing the solvent, solvent compressor **130** is configured to introduce the pressurized solvent into detachable canister system **140**.

(24) As FIG. **1** shows, detachable canister system **140** is connected in fluid communication with solvent compressor **130**. As FIG. **1** illustrates, detachable canister system **140** includes a plurality of detachable canisters, including a first detachable canister **150**, a second detachable canister **163**, and a third detachable canister **164**. Detachable canister system **140** is configured to direct solvent from solvent compressor **130** to each detachable canister via a detachable canister line **181**.

(25) Detachable canister system **140** is configured to fluidly receive compressed solvent from solvent compressor **130**. Detachable canister system **140** is further configured to direct to extract container **170** extract solution produced within attached canisters, the extract solution including both solvent and solute extracted from source materials contained in the canisters. Detachable canister system **140** is further configured to direct to extract container **170** any overflow solvent output by solvent compressor **130** and not received by a detachable canister.

(26) As FIG. **1** illustrates, detachable canister system **140** is configured to direct fluid from solvent compressor **130** to each detachable canister. As FIG. **1** shows, detachable canister system **140** includes an input valve associated with each detachable canister. Each input valve controls fluid communication between solvent compressor **130** and the associated detachable canister. When an input valve is opened, solvent compressor **130** is configured to communicate contained compressed solvent to the associated canister.

(27) As FIG. **1** shows, solvent compressor **130** is configured to communicate with each detachable canister individually. Accordingly, detachable canister system **140** allows a user to refill a selected canister as one or more of the other canisters remain closed and to continue extracting solute from contained source material.

(28) As FIG. **1** shows, detachable canister system **140** is configured to direct the extract solution created in each detachable canister to extract container **170** via an extract mixture line **184**. As FIG. **1** shows, detachable canister system **140** includes an output valve associated with each detachable canister. When an output valve is opened, the associated canister is placed in fluid communication with extract mixture line **184**.

(29) When placed in fluid communication, the associated canister is configured to output an extract mixture to extract container **170** via extract mixture line **184**. A user may use the output valves to direct the extract mixture contained in an associated canister to extract container **170**. In some examples, first pump **101** and second pump **102** are configured to cooperatively suck the extract mixture from the associated canister toward extract container **170**.

(30) As FIG. **1** illustrates, detachable canister system **140** additionally includes an overflow line **182** in fluid communication with solvent compressor **130**, each detachable canister, and extract container **170**. Overflow line **182** is configured to direct overflow solvent that does not make it from solvent compressor **130** to one of the detachable containers after charging. For example, overflow line **182** may be used to collect solvent trapped in detachable canister line **181** after filling one of the detachable canisters with solvent.

(31) FIG. **2** illustrates an example detachable canister, first detachable canister **150**, filled with solvent and source material. In FIG. **2**, first detachable canister **150** is currently extracting solute

from the source material. As FIG. 2 illustrates, first detachable canister **150** includes a top portion **154**, which may be screwingly attached to and detached from a bottom portion **156**. When top portion **154** and bottom portion **156** are attached, they define a fluid-tight container configured to store solvent and source material during extraction.

(32) As FIG. 2 shows, first detachable canister **150** may additionally or alternatively include a top mesh filter **149** and a bottom mesh filter **151**. As FIG. 2 shows, top mesh filter **149** defines a perimetral gasket surrounding a mesh filter. The perimetral gasket is made of a food-grade nitrile, allowing first detachable canister **150** to be legally used to extract solute that may be used for food products. As FIG. 2 illustrates, top mesh filter **149** is configured to be slidably inserted into the top of first detachable canister **150** to prevent sediment from inadvertently backflowing through the input of first detachable canister **150**. The perimetral gasket is sized to partially compress within first detachable canister **150**, thus frictionally supporting top mesh filter **149** in a substantially fixed position within first detachable canister **150**.

(33) In some examples, the filter of top filter **149** may define a stainless steel **200** mesh filter. In some examples, the gasket portion of top filter **149** may define food grade nitrile.

(34) Bottom mesh filter **151** is substantially similar to top mesh filter **149**, but is positioned proximate the bottom of first detachable canister **150**. Accordingly, bottom mesh filter **151** is configured to prevent sediment from inadvertently flowing through the output of first detachable canister **150**.

(35) Because system **100** includes multiple canisters and each canister is removable, system **100** is able to extract solute in the connected, filled containers as other functions of system **100** continue to operate. For example, when one canister is attached and extracting, solvent compressor **130** may charge solvent for a second canister. Further, a user may be able to load a detached canister as solvent compressor **130** charges solvent, providing even greater parallelism.

(36) As FIG. 2 shows, detachable canister system **140** includes, for each detachable canister, both an upper canister attachment device **153** and a lower canister attachment device **159**. Each upper canister attachment device **153** and lower canister attachment device **159** is configured to selectively secure the associated detachable canister. As shown in FIG. 2, each upper canister attachment device **153** device includes a handle **155**, which defines an over-center securing lever configured to be pulled to lock upper canister attachment device **153** in a substantially fixed position over the top opening of the associated canister. As FIG. 2 shows, each upper canister attachment device additionally defines a compressible, fluid tight gasket **157** configured to be compressed against the associated canister when upper canister attachment device **153** is locked in a closed configuration. As FIG. 2 shows, upper canister attachment device **153** may be pulled away and spaced from the associated canister when handle **155** being released.

(37) As FIG. 2 shows, lower canister attachment device **159** is configured to slidably receive the lower opening of an associated canister. As FIG. 2 illustrates, lower canister attachment device **159** includes a compressible, fluid tight gasket **157**. As FIG. 2 illustrates, a user may manipulate upper canister attachment device **153** to receive the associated canister such that the canister is engaged with both gaskets **157**. When a user pulls handle **155** to position upper canister attachment device **153** in a locked configuration, both gaskets **157** are configured to compress to place the canister in fluid communication with both solvent compressor **130** and extract container **170**. Similarly, both gaskets are configured to release when handle **155** is released. When handle **155** is released and upper canister attachment device **153** is pulled away from the associated canister, the canister can be slidably removed from lower canister attachment device **159** to be removed from system **100**.

(38) As FIG. 1 illustrates, each detachable canister includes a heating pad **152** wrapped around its exterior. Each heating pad **152** is configured to receive electrical energy, such as by being plugged into electrical outlet **89**, to heat the canister around which it is wrapped. Increasing the temperature can, in many cases, increase solvents' efficacy and efficiency in extracting solute from a source material. The heating pads may be controlled by an electronic heating pad controller **78**, which

configures the heating pad to operate at a chosen temperature or intensity.

(39) As FIG. 1 shows, extract container **170** is in fluid communication with detachable canister system **140**, configured to receive extract solution output by the detachable canisters and any overflow solvent transmitted by overflow line **182**. As FIG. 4 illustrates, extract container **170** includes a container input **172**, a container output **174**, and a lid **176**. Extract container **170** is configured to receive the extract mixture output by the detachable canisters. Extract container **170** is further configured to separate post-extraction solvent from the extract mixture and output the post-extraction solvent for reclamation.

(40) Container input **172** is configured to receive extract mixture in a liquid state from the detachable canisters. For example, FIG. 4 illustrates extract container **170** containing a liquid **72** containing both extract mixture and extracted essential oil. The extract mixture has been collected from detachable canister system **140** in the current cycle of system **100**, whereas the extracted essential oil is the residual essential oil after reclaiming post-extraction solvent from a previous cycle of system **100**.

(41) As FIG. 4 shows, system **100** includes features that are configured to restrict fluid from passing back into detachable canister system **140**. As FIG. 4 shows, container input **172** is positioned below the midpoint of extract container **170**. At this point, container input **172** will often be positioned below collected essential oils. Because container input **172** is often positioned within collected liquid, extract container **170** directs gas, such as evaporated post-extraction solvent **71**, toward container output **174** rather than the submerged container input **172**.

(42) Further, container input **172** includes angled open ends **173**. The open ends **173** prevent liquid from being directed toward lid **176**. By preventing liquid from contacting lid **176**, extract container **170** provides a substantially clear view of the liquid contained in extract container **170**. Further pump **102** and pump **104** are configured to direct fluid toward container output **174** and away from container input **172**.

(43) As FIG. 4 shows, container output **174** is configured to direct gas, such as evaporated post-extraction solvent **71**, from extract container **170** to solvent collection container **115**. As FIG. 3 illustrates, container output **174** is positioned above container input **172** and above the top of fluid collected in extract container **170**. Because container output **174** is located in this elevated position, it is positioned to receive gas from extract container **170** as fluid remains in extract container **170**. Because container output **174** is positioned to primarily receive gas from extract container **170**, extract container **170** is configured to separate evaporated post-extraction solvent from extract mixture while leaving the extracted essential oils in extract container **170**. This results in a pure product while reclaiming post-extraction solvent at a high rate.

(44) Extract container **170** defines a pressure pot, configured to retain its structure at a wide range of pressure profiles. Namely, extract container **170** is configured to maintain its structure from -30 mmHg of vacuum pressure to 300 pounds per square inch of positive pressure. In typical working conditions, the amount of pressure applied to container **170** will range from -30 mmHg of vacuum pressure to 150 pounds per square inch of positive pressure. In particular, extract container **170** will often be between -30 to $\bigcirc$ mmHg of vacuum when receiving fluid from extract mixture line **184** and between $\bigcirc$ and 60 pounds per square inch of pressure when directing fluid to container output **174**.

(45) As FIGS. 3 and 4 show, lid **176** is detachably secured to the top of extract container **170**. Lid **176** allows a user to view the contents within container **170** and is made from plexiglass. The thick Plexiglas construction of lid **176** provides sufficient rigidity and structural integrity to withstand the widely disparate pressure conditions often present in extract container **170**. The lid may be made from any material configured to withstand anticipated operating pressures.

(46) As FIGS. 3 and 4 show, lid **176** is fastened to the main body of container **170** by a series of half-inch bolts **177**. The bolts are each detachable, allowing lid **176** to be selectively removed. Further, the numerosity and strength of the bolts provide sufficient structural support to restrict lid

176 from being damaged or unintentionally removed under the pressure conditions typically encountered during operation.

(47) As FIG. 3 shows, lid **176** includes ports through which container output **174** and container input **172** are spaced at a distance selected to retain structural integrity of lid **176** under the pressure conditions typically encountered during operation.

(48) As FIG. 1 illustrates, an extract container heating element **178** is thermally coupled with the contents of extract container **170**, being positioned at the bottom of extract container **170**. Heating element **178** defines an electrically powered heating pad rated at 500 Watts. Heating pad **152** is configured to heat the extract mixture to a distilling temperature to produce an evaporated portion of the solvent in extract container **170**. The distilling temperature to which extract container **170** is heated is greater than the boiling point of butane and less than typical essential oil boiling points.

(49) Heating element **178** may be powered by an electrical connection to electrical outlet **89**.

Additionally, the temperature or intensity of heating element **178** may be controlled by an electronic container heating element controller **79**.

(50) As FIG. 1 illustrates, system **100** includes a pressure release valve **185** and pressure release line **187**, both in fluid communication with container output **174**. Pressure release line **187** is in fluid communication with ambient air at a location in which it is safe to output flammable fluids. During operation, the pressure of the interior of extract container **170** may fall outside the bounds of desirable operating pressures. For example, pressure release valve **185** is configured to open pressure release line **187** upon extract container **170** exceeding 150 pounds per square inch of positive pressure or negative 30 pounds per square inch of vacuum pressure.

(51) As FIG. 1 illustrates, first pump **101** and second pump **102** are connected in fluid communication with extract container **170**. First pump **101** and second pump **102** are configured to direct fluid through system **100** in the direction indicated by arrows **98** and **99** shown in FIG. 1. For example, first pump **101** and second pump **102** are configured to cooperate to direct evaporated post-extraction solvent from extract container **170** to solvent collection container **115** over a container output line **188** and a solvent collection line **190**. In some examples, a fan may be attached between first pump **101** and second pump **102** for cooling when system **100** is operational.

(52) Because system **100** defines a closed loop system, first pump **101** and second pump **102** are cooperatively configured to direct fluid at a variety of stages of system **100** as long as any intervening valves are opened. For example, first pump **101** and second pump **102** may be configured to cooperate to draw solvent from solvent collection container **115** to solvent source container **120**. Further, first pump **101** and second pump **102** are configured to cooperatively communicate at least a portion of post-extraction solvent evaporated within extract container **170** to solvent collection container **115**.

(53) Similarly, first pump **101** and second pump **102** may be configured to cooperatively direct extract solution output from the detachable canisters to extract container **170**. In some examples, backflow pressure produced by first pump **101** and second pump **102** provide some or all of the pressure used by solvent compressor **130** to pressurize solvent prior to introducing it to the solvent containers.

(54) First pump **101** and second pump **102** collectively produce a flow rate sufficient to accomplish each of the aforementioned functionalities. Some examples include more or fewer pumps connected in series. Additional pumps may provide additional pumping power, whereas fewer pumps may save on operating costs.

(55) As FIG. 1 shows, solvent collection line **190** is configured to pass through condensing system **105** between second pump **102** and solvent collection container **115**. As FIG. 1 illustrates, condensing system **105** includes a coolant pump **104**, a first condenser column **107**, a second condenser column **108**, a coolant line **109**, a coolant loop line **113**, a first expansion valve **111**, and a second expansion valve **112**. Condensing system **105** is configured to cool post-extraction solvent pumped by first pump **101** and second pump **102** prior to the solvent reaching solvent collection

container **115**. By cooling the post-extraction solvent, condensing system **105** allows the post-extraction solvent to be stored and collected as a liquid.

(56) Coolant line **109** includes a coolant defining a refrigerant selected to cool when evaporated. As FIG. **1** illustrates, coolant pump **104** is configured to direct the coolant contained in coolant line **109** in the direction indicated by arrow **96** and arrow **97**.

(57) As FIG. **1** illustrates, coolant line **109** is routed through second condenser column **108**. After passing through second condenser column **108**, coolant line **109** splits, with one branch being directed back to coolant pump **104** and the other branch being directed toward first condenser column **107**. As FIG. **1** shows, coolant line **109** joins coolant loop line **113** in fluid communication after passing through second expansion valve **112**.

(58) Coolant line **109** is configured to absorb heat from the post-extraction solvent passing through solvent collection line **190** to cool the post-extraction solvent to a liquid state. Coolant line **109** and coolant loop line **113** additionally cooperate to continuously cool coolant pump **104** during operation.

(59) As FIG. **5** illustrates, solvent collection line **190** defines a coiled portion **114** through first condenser column **107**. Likewise, coolant line **109** defines a coiled portion **116** through first condenser column **107**. As FIG. **5** shows, coiled portion **116** and coiled portion **114** are sufficiently close with one another for coolant line **109** and solvent collection line **190** to be thermally coupled through first condenser column **107**. As FIG. **5** shows, first condenser column **107** is filled with a thermally conductive liquid **106**, such as an antifreeze, which encourages rapid heat transfer between coolant line **109** and solvent collection line **190**.

(60) As FIG. **1** shows, coolant line **109** is connected in fluid communication with coolant loop line **113**. Coolant loop line **113** is routed through second condenser column **108** and defines a coiled portion through second condenser column **108**. Similar to the design of first condenser column **107**, the coiled portion of coolant loop line **113** and a second coiled portion of coolant line **109** are engaged and thermally coupled with one another within second condenser column **108**.

(61) As FIG. **1** shows, the hydraulic circuit defined by coolant line **109** and coolant loop line **113** routes coolant pumped by coolant pump **104** through two expansion valves. Coolant passing through first expansion valve **111** in second condenser column **108** expands to vapor or a vapor/liquid mix, which draws heat from the second coiled portion of coolant line **109**. This cools the coolant line **109**, particularly at its second portion. As FIG. **1** shows, a portion of this cooled coolant is reintroduced into coolant pump **104**, thereby continuously cooling coolant pump **104** during operation.

(62) Coolant passing through second expansion valve **112** as it returns to coolant pump **104** similarly expands to a vapor or vapor/liquid mix, which draws heat from solvent collection line **190** and the solvent passing therein. In many cases, second expansion valve **112** will expand coolant returning from coolant line **109**, which draws heat from solvent collection line **190**. This cools the post-extraction solvent passing through solvent collection line **190** to a liquid. Cooling the solvent and shifting it to a liquid decreases losses of recaptured post-extraction solvent and increases the efficiency of solvent collection container **115** in storing previously used solvent, thus increasing the recapture rate of system **100**.

(63) The coiling of fluid lines through both first condenser column **107** and second condenser column **108** increases the surface area throughout which the corresponding lines are thermally coupled. By maximizing this surface area, first condenser column **107** and second condenser column **108** are better able to transfer heat between the two paired lines. Further, the coiling of the lines increases the amount of time in which contained fluids are exposed to one another, further increasing the columns' cooling efficiency.

(64) As FIG. **1** illustrates, solvent collection container **115** is connected in fluid communication with extract container **170** through solvent collection container **115** and solvent source container **120** via a storage container line **194**. As FIG. **1** shows, solvent collection container **115** includes a

collection container input **117** and a collection container output **118**. Solvent collection container **115** is configured to collect post-extraction solvent separated from the distilled solute within extract container **170** and directed through solvent collection line **190**.

(65) Solvent collection container **115** is configured to direct collected solvent to solvent source container **120** upon collecting a maximum amount of collected solvent. In some configurations, solvent collection container **115** is configured to communicate collected solvent upon exceeding its storage capacity. In other examples, solvent collection container **115** is configured to communicate collected solvent upon the collected solvent reaching the end of collection container output **118**. When the collected solvent reaches the second storage unit output, the suction force produced by first pump **101** and second pump **102** draws collected solvent through the second storage unit output to refill solvent source container **120**.

(66) Solvent collection container **115**, solvent source container **120**, and extract container **170** define volumes that are operatively paired with one another. Extract container **170**, solvent source container **120**, and solvent collection container **115** may, at times of operation, be configured to receive all of the fluid from the preceding fluidly connected elements. Accordingly, solvent collection container **115** and solvent source container **120** are sized to ensure sufficient storage space for any post-extraction solvent output by extract container **170**.

(67) To ensure sufficient headroom, the combined volume of solvent source container **120** and solvent collection container **115** may be equal to the volume of extract container **170**. Likewise, extract container **170** may define a volume equal to the combined volume of solvent source container **120** and solvent collection container **115** to provide sufficient headroom to be filled with all of the solvent initially contained by solvent collection container **115** and solvent source container **120**. In some examples, one or more of solvent source container **120**, solvent collection container **115**, and extract container **170** may be sized larger than needed, which may guarantee sufficient headroom.

(68) Although solvent collection container **115** and solvent source container **120** are distinct containers in system **100**, this disclosure contemplates that a single container could serve as both a solvent collection container and solvent source container.

(69) Turning attention to FIG. **6**, a second example of a system for extracting solute from a source material, system **200**, will now be described. System **200** includes many similar or identical features to system **100**. Thus, for the sake of brevity, each feature of system **200** will not be redundantly explained. Rather, key distinctions between system **200** and system **100** will be described in detail and the reader should reference the discussion above for features substantially similar between the two systems.

(70) As FIG. **6** shows, some example systems for extracting solute from a source material may include more than one extract container. For example, system **200** includes both a first extract container **270a** and a second extract container **270b**. It will be appreciated that second extract container **270b** has a substantially similar configuration to first extract container **270a**, and each of extract containers **270a** and **270b** have a substantially similar configuration to extract container **170** (i.e., having a container inputs **272a** and **272b**, a container outputs **274a** and **274b**, a lids **276a** and **276b**, and a heating elements **278a** and **278b**).

(71) In this example, second extract container **270b** is coupled to output line **288** downstream of first extract container **270a** and is in fluid communication with first extract container **270a**. In other words, first extract container **270a** is coupled to detachable canister system **240** via overflow line **282** for collection of overflow solvent and extract mixture line **284** for collection of the extract mixture, while second extract container **270b** is coupled to first extraction container **270a** for collection of the extract mixture. Further, in this example, overflow line **282** and extract mixture line **284** are both single pathway lines. Alternatively, the second extract container **270b** can be coupled directly to the detachable canister system.

(72) As FIG. **6** illustrates, system **200** further includes additional solvent collection containers,

compared to system **100**, operatively paired with the two extract containers of system **200**. Namely, system **200** includes a first solvent collection container **215**, a solvent collection container **216**, and a solvent collection container **217**. System **200** additionally includes a solvent source container **220**. The combined volume of first solvent collection container **215**, solvent collection container **216**, solvent collection container **217**, and solvent source container **220** is operatively paired with the combined volume of first extract container **270a** and second extract container **270b**.

(73) With reference to FIG. 7, an example of a method for extracting solute from a source material, method **300**, will now be described. Some of the steps of method **300** may be carried out using system **100**, or other disclosed systems. Accordingly, the discussion of method **300** will reference system **100**. Although this disclosure references system **100** in connection with method **300**, method **300** is not required to be carried out on equipment similar to system **100**, system **200**, or other disclosed systems.

(74) As FIG. 7 illustrates, method **300** includes depositing a source material including a solute in a canister at step **305**, removably attaching the canister in fluid communication with an extract container at step **310**, introducing a solvent into the canister at step **315**, and exposing the source material to the solvent for a predetermined period of time at step **320**, and communicating the extract mixture to an extract container at step **325**. As FIG. 7 shows, method **300** further includes separating the recycled solvent at step **330**, cooling the recycled solvent at step **335**, collecting the recycled solvent in a solvent collection container at step **340**, and introducing at least a portion of the recycled solvent from the solvent collection container to the canister at step **345**.

(75) As FIG. 7 illustrates, a source material including a solute is deposited in a canister at step **305**. In some examples, the source material may define plant material from which essential oils may be extracted. For example, FIG. 2 illustrates first detachable canister **150** extracting essential oils from contained lavender **91**.

(76) In some examples, source material is deposited in a substantially fluid-tight canister configured to selectively receive solvent and selectively communicate extract mixture created therein. For example, first detachable canister **150** is substantially fluid tight in a closed configuration, but includes an input and output allowing fluid to be selectively input solvent and output created extract mixture.

(77) In some examples, canisters may be configured to be detachably connected to an extraction system. In some such examples, source material may be deposited detachable canisters when detached from its associated extraction system. For example, first detachable canister **150**, as shown in FIG. 2, defines a selectively openable canister configured to be selectively attached and detached from system **100**. Accordingly, first detachable canister **150** allows a user to load it with source material when it is detached from system **100**.

(78) Some examples include a plurality of detachable canisters that may be individually attached and detached from extraction systems. In such examples, source material may be placed in one or more of the canisters while the other canisters continue normal operation. For example, system **100** includes three detachable canisters, first detachable canister **150**, second detachable canister **163**, and third detachable canister **164**. First detachable canister **150**, for example, could be removed and filled with source material as system **100** continues the extraction/reclamation process with second detachable canister **163** and third detachable canister **164**.

(79) Some examples may include a detachable canister interface allowing canisters to be easily detached and reattached. In some examples, detachable canister interfaces are configured to receive the canister to support the canister in fluid communication with the solvent source container and an extract container. For example, system **100** includes detachable canister system **140** that allows each of the detachable canisters to be attached and detached. As FIG. 1 shows, each of first detachable canister **150**, second detachable canister **163**, and third detachable canister **164** are placed in fluid communication with solvent source container **120** and extract container **170** when attached.

(80) In some examples, detachable canister interfaces are configured to release the canister in response to user manipulation. In some examples, such as system **100**, detachable canister interfaces allow users to attach and detach canisters without any specific tools or hardware.

(81) As FIG. 7 shows, the canister containing the source material is removably attached in fluid communication with an extract container at step **310**. By placing the canister in fluid communication with an extract container, the canister is able to communicate created extract mixture to the extract container. The extract container may then be used to collect extracted essential oils and separate recycled solvent from extracted essential oils collected therein.

(82) As previously discussed, some examples include a plurality of simultaneously attached canisters. In some such examples, two or more of the simultaneously attached canisters may simultaneously be in fluid communication with a connected extract container. By placing each canister in fluid communication with the extract container, one or more canisters may simultaneously output contained extract mixture to a single connected extract container.

(83) As seen in FIG. 7, solvent is introduced into the canister at step **315**. In some examples, solvent is introduced into the canister by communicating, fluidly, solvent from a solvent source container to the canister. As previously discussed, some examples may include a plurality of detachable canisters. This disclosure contemplates introducing solvent into each canister independently, simultaneously with introducing solvent into one or more of the other canisters.

(84) In some examples, introducing solvent into the canister includes pressurizing a charging portion of the solvent prior to introducing the solvent to the canister. In some examples, a charging portion of solvent may be pressurized when contained in a solvent compressor as one or more of the attached canisters contain an extracting portion of solvent being used to extract solute from the source material. In system **100**, for example, a user may pressurize a charging portion of solvent in solvent compressor **130**, designated for first detachable canister **150**, as second detachable canister **163** and third detachable canister **164** each contain an extracting portion of solvent and are extracting solute therewith.

(85) Pressurizing the solvent is often a time-consuming process. Extracting solute in the canisters is also often time consuming. As a result, the parallelism afforded by pressurizing solvent as other attached canisters continue the extraction process efficiently streamlines the pressurizing and extracting steps of disclosed methods.

(86) In some examples, some or all of the solvent introduced into the canister may include recycled solvent reclaimed from a previous cycle of the disclosed methods. As will be discussed more below, some examples include a reclamation methodology that operates alongside disclosed extraction methodologies. By using reclaimed solvent, disclosed methods may use solvent particularly efficiently. In some examples, solvent may automatically be collected and reintroduced. In some examples, solvent collection and reintroduction may occur simultaneously with other steps of the disclosed methods. Some examples may collect solvent in a plurality of solvent collection containers, such as system **200**.

(87) In some examples, introducing the solvent may include passing the solvent through a solvent filter as it passes from solvent source container **120** to solvent compressor **130**. In some examples, the solvent filter may define a 13-X molecular sieve configured for membrane filtration of the solvent as it passes from solvent source container **120** to solvent compressor **130**.

(88) As FIG. 7 illustrates, the source material is exposed to the solvent for a predetermined period of time at step **320**. The predetermined period of time in which the source material exposed is selected to substantially maximize the purity of extracted solvent. In some examples, the source material and solvent are exposed to heat and pressure conditions that may increase the efficiency with which solute is extracted from source materials. Soaking the source material in the solvent within an associated canister for 3-5 minutes has been found to be a surprisingly effective method exposing the source material to the solvent. In some examples, following the soak with a hydrocarbon wash of the associated canister has been found to result in a particularly high quality,

pure product in subsequent extraction steps performed with that particular canister.

(89) For example, it may be desirable to extract solute from source materials at both high temperatures while solvent remains in a liquid state. This disclosure contemplates both heating the solvent when it is exposed to the source material and pressurizing the contained volume of solvent to a selected pressure to maintain the solvent in a liquid state when heated. By manipulating the pressure and temperature of the solvent, disclosed methods may extract solute at a higher purity and greater yield per unit of source material than conventional extraction methods.

(90) As shown in FIG. 7, the created extract mixture is communicated to an extract container, the extract container in fluid communication with the extract mixture, at step **325**. By communicating the extract mixture to the extract container, the extract container may collect and store extract for future use. In some examples, extract containers may be removed to use or store collected extract.

(91) In some examples, the extract container may define a negative pressure prior to receiving extract mixture. The created extract mixture may be communicated to the extract container by opening a fluid communicative path between a canister containing created extract mixture and the extract container. For example, system **100** allows a user to manipulate the output valve of detachable canister system **140** associated with a container containing created extract mixture to open a fluid line between the associated canister and extract container **170**.

(92) As FIG. 7 shows, the recycled solvent is separated at step **330**. Separating the recycled solvent may include heating the container to evaporate the recycled solvent. Heating the container to evaporate the recycled solvent may include heating the container to a solvent extraction temperature. The solvent extraction temperature of the container may be greater than a boiling point of the solvent and less than a boiling point of the solute. By raising the temperature of the extract mixture above the solvent's boiling point and below the boiling point of essential oils, the solvent is separated from the extract mixture as a gas. The evaporated solvent may output independently of any contained essential oil or other extract.

(93) In some examples, users may discard the contents of an extract container after evaporating the recycled solvent. In some examples, the contents of the extract container may include odorants or other impurities that may remain in the extract container after evaporating the recycled solvent. As a result, the recycled solvent may have increased purity compared the input solvent. Users may discard these impurities to ensure that they do not end up in any end product produced by subsequent extraction/reclamation cycles.

(94) In some examples, the solvent may define butane. In such examples, heating the container to evaporate the solvent may include raising the temperature within the container above butane's sea level boiling point of about 30.8 degrees Fahrenheit while maintaining the temperature within the container to below typical boiling points of water and/or essential oils.

(95) In some examples, separating the recycled solvent includes receiving the evaporated recycled solvent through a container output opening. In some examples, one or more fluidly-connected pumps may suck evaporated recycled solvent through the container output opening. As previously discussed, this disclosure contemplates extract containers that remain structurally stable at negative pressures. Attached pumps may be configured to extract substantially all of the evaporated recycled solvent contained in an extract container and leave the extract container with a negative pressure. By leaving the extract container with a negative pressure, the pumps additionally prepare the extract container to later receive additional extract mixture from one or more attached canisters.

(96) In some examples, the evaporated recycled solvent is positioned above any liquid extract mixture contained in the extract container. For example, extract container **170** includes container output **174** positioned near the top of extract container **170** and above any contained extract. In some examples, users may periodically empty the extract to ensure that the container output opening remains above any contained extract. For example, the extract container may be removed and contained extract may be stored in an alternative container. To ensure the container output opening remains above the contained extract, the container may be periodically emptied prior to the

extract container accumulating sufficient extract to reach the container output opening.

(97) As seen in FIG. 7, the recycled solvent is cooled at step 335. In some examples, as is seen in system 100, the recycled solvent is thermally coupled with a coolant for a portion of the time after the recycled solvent leaves an extract container. In some examples, the recycled solvent is cooled prior to collecting the recycled solvent in the solvent collection container.

(98) In some examples, the recycled solvent is directed through a solvent collection line and the recycled solvent is thermally coupled with a coolant along at least a portion of the solvent collection line. In some examples, the coolant is directed through a coolant line which is engaged with the solvent collection line over at least a portion of its length. For example, coolant line 109 is configured to carry coolant and is engaged with solvent collection line 190 through first condenser column 107. Because solvent collection line 190 is engaged with coolant line 109 and each line is constructed of thermally conductive material through first condenser column 107, coolant line 109 is thermally coupled with solvent collection container 115 through first condenser column 107.

(99) In some examples, the coolant is directed through a fluid-transmissive coolant loop. The coolant loop may be configured to both output and receive coolant from the coolant line. For example, FIG. 1 illustrates an example coolant loop line, coolant loop line 113, which is configured to output and receive coolant from the coolant line at a single junction.

(100) The coolant loop may additionally or alternatively define an expansion valve configured to expand and cool the coolant in the coolant loop. Additionally, or alternatively, the coolant loop line may be engaged with the coolant line over at least a portion of its length, thereby thermally coupling the coolant loop and the coolant line over a portion of their lengths. By expanding the coolant in the coolant loop, the coolant loop includes a coolant that may be cooler than the coolant in the primary coolant line. By thermally coupling the coolant loop with the coolant line, the coolant loop may, in effect, cool the coolant in the coolant line.

(101) FIG. 1 illustrates an example fluid circuit including a coupled coolant loop and coolant line. As previously discussed, coolant line 109 is thermally coupled with solvent collection line 190. As FIG. 1 shows, coolant line 109 is connected to both input and output to coolant loop line 113 at a single junction. Accordingly, system 100 is configured to allow coolant to be directed through coolant line 109 and is configured to direct at least a portion of the coolant in the coolant line 109 through coolant loop line 113.

(102) As FIG. 1 shows, coolant loop line 113 includes first expansion valve 111 which is configured to expand and cool coolant contained in coolant loop line 113 prior to directing coolant in coolant loop line 113 through second condenser column 108. As previously discussed, coolant loop line 113 is thermally engaged with coolant line 109 through second condenser column 108. By expanding coolant contained in coolant loop line 113 and thermally coupling coolant loop line 113 and coolant line 109 immediately downstream of this expansion, coolant loop line 113 is configured to cool coolant line 109 through second condenser column 108.

(103) As illustrated in FIG. 7, the recycled solvent is collected in a solvent collection container in fluid communication with the extract container at step 340. In some examples, the solvent collected by the solvent collection container is fluidly communicated from an extract container. In some examples, the recycled solvent is cooled to a liquid state prior to collection in the solvent collection container.

(104) In some examples, collected solvent is automatically output from a solvent collection container upon the solvent collection container collecting a maximum amount of collected solvent. In some examples, the automatically output collected solvent is input into a solvent source container in fluid communication with the associated solvent collection container. For example, solvent collection container 115 is configured to output collected recycled solvent to solvent source container 120 upon collecting a maximum quantity of collected recycled solvent.

(105) In some configurations, solvent collection containers are configured to communicate collected solvent upon exceeding its storage capacity. In other examples, solvent collection

containers are configured to automatically communicate collected solvent upon the collected solvent reaching the end of a collection container output positioned within the solvent collection container.

(106) In some examples, collecting the recycled solvent includes displacing, with a pump, evaporated recycled solvent from the extract container to the solvent collection container. For example, first pump **101** and second pump **102** are configured to, in certain configurations, draw collected solvent from extract container **170** to solvent collection container **115**.

(107) In some examples, collecting the recycled solvent includes displacing, with a pump, recycled solvent from solvent collection containers to solvent source containers or canisters. For example, first pump **101** and second pump **102** are configured to, in certain configurations, draw collected solvent from solvent collection container **115** to solvent source container **120**. First pump **101** and second pump **102** may be further configured to draw reclaimed solvent in solvent source container **120** to one or more connected detachable canisters containing source material.

(108) In some examples collecting the recycled solvent includes sealing the solvent collection container when the solvent collection container contains at least a portion of the recycled solvent and detaching the solvent collection container. In some examples, the solvent collection container may be sealed when it contains a predetermined quantity of the recycled solvent. Upon being sealed, solvent source containers containing reclaimed solvent may be detached and stored for later use. By allowing removal and storage of solvent collection containers and/or solvent source containers, users may store purified, reclaimed solvent for use in future use. Because of the disclosed purification features, recycled solvent may be of a greater purity than many commercially available solvents.

(109) In some examples, recycled solvent is collected in an additional solvent collection container in fluid communication with the extract container. Additional solvent collection containers may be useful, for example, when additional or larger extract containers are used, as they may provide the increased headroom required when using additional or larger extract containers. In some such examples, the second solvent container may be operatively paired with the extract container. In some examples, solvent containers and extract containers may be operatively paired by defining a substantially similar total solvent container volume that is consistent with or equal to the total extract container volume. System **200**, for example, includes a supplemental extract container paired with a supplemental solvent collection container.

(110) As shown in FIG. 7, recycled solvent from the solvent collection container is introduced into the canister at step **345**. In some examples, introducing at least a portion of the recycled solvent from the solvent collection container to the canister includes directing at least a portion of the recycled solvent from the solvent collection container to a solvent source container prior to reaching a canister. By reintroducing recycled solvent, disclosed methods efficiently re-use reclaimed solvent from previous extraction/reclamation cycles. Because used solvent is not simply discarded, disclosed methods provide robust economic and ecological efficiency.

(111) In some examples, recycled solvent contained in solvent collection container is displaced into the solvent source container upon the solvent collection container collecting a predetermined quantity of recycled solvent. In some examples, the solvent collection container is configured to introduce such received recycled solvent to the canister in future cycles, instead of adding additional solvent. For example, solvent source container **120** is configured to direct new solvent and/or solvent received from solvent collection container **115** to detachable canister system **140**.

(112) The disclosure above encompasses multiple distinct inventions with independent utility. While each of these inventions has been disclosed in a particular form, the specific embodiments disclosed and illustrated above are not to be considered in a limiting sense as numerous variations are possible. The subject matter of the inventions includes all novel and non-obvious combinations and subcombinations of the various elements, features, functions and/or properties disclosed above and inherent to those skilled in the art pertaining to such inventions. Where the disclosure or

subsequently filed claims recite “a” element, “a first” element, or any such equivalent term, the disclosure or claims should be understood to incorporate one or more such elements, neither requiring nor excluding two or more such elements.

(113) Applicant(s) reserves the right to submit claims directed to combinations and subcombinations of the disclosed inventions that are believed to be novel and non-obvious. Inventions embodied in other combinations and subcombinations of features, functions, elements and/or properties may be claimed through amendment of those claims or presentation of new claims in the present application or in a related application. Such amended or new claims, whether they are directed to the same invention or a different invention and whether they are different, broader, narrower or equal in scope to the original claims, are to be considered within the subject matter of the inventions described herein.

Claims

1. A system for extracting solute from a plant material, the system comprising: a solvent source positioned to receive and provide a solvent that includes at least a hydrocarbon solvent; a canister adapted to contain the plant material, the canister positioned downstream of the solvent source for receiving the solvent, thereby producing an extract solution of the solvent and an essential oil extracted from the plant material; an extract container positioned for receiving and collecting at least a portion of the extract solution from at least the canister, the extract container further positioned to separate at least a portion of the solvent from at least the portion of the extract solution, thereby providing a recycled solvent; a solvent collection line for directing the recycled solvent, the solvent collection line at least partially defining a fluid communication path between the extract container and the solvent source; and a coolant line positioned for directing coolant that is thermally coupled to at least a portion of the solvent collection line for cooling the recycled solvent that is flowing through the solvent collection line, with at least one of the solvent collection line and the coolant line defining a coiled portion along the portion of the solvent collection line for cooling the recycled solvent.
2. The system of claim 1, with the solvent collection line defining at least a first pipe that is positioned for directing, in a first direction along the solvent collection line, the recycled solvent from the extract container and towards the solvent source; and the coolant line that defines at least a second pipe positioned for directing the coolant that is thermally coupled to at least the portion of the solvent collection line for cooling the recycled solvent that is flowing through the solvent collection line in the first direction, with at least one of the first pipe and the second pipe defining the coiled portion.
3. The system of claim 1, with the solvent collection line thermally coupled with the coiled portion of the coolant line.
4. The system of claim 1, with the coolant line thermally coupled with the coiled portion of the solvent collection line.
5. The system of claim 1, with the solvent collection line comprising a first coiled portion and the coolant line comprising a second coiled portion, the first coiled portion thermally coupled to the second coiled portion.
6. The system of claim 1, further comprising a solvent collection container positioned for receiving, collecting, and providing the recycled solvent, the solvent collection container further positioned downstream of the solvent collection line and upstream of the solvent source.
7. The system of claim 1, further comprising a pump positioned for at least directing the recycled solvent through the solvent collection line, the pump further positioned downstream of the extract container and upstream of at least the portion of the solvent collection line for cooling the recycled solvent.
8. The system of claim 1, further comprising: canisters that each contain the plant material, each of

the canisters positioned in selective fluid communication with the solvent source for receiving the solvent; and a canister interface configured to support the canisters.

9. The system of claim 8, with the canister interface comprising: an input valve for an associated canister of the canisters, the input valve operable to selectively adjust between at least an open position for allowing fluid to enter the associated canister and a closed position for restricting fluid from entering the associated canister; and an output valve for the associated canister, the output valve operable to selectively adjust between at least an open position for allowing fluid to leave the associated canister and a closed position for restricting fluid from leaving the associated canister.

10. The system of claim 8, further comprising a heating element thermally coupled with an associated canister of the canisters, the heating element operable to apply heat to the associated canister.

11. The system of claim 10, with the heating element operable to apply heat to the associated canister of at least a distilling temperature, the distilling temperature being at least a boiling point of the solvent.

12. The system of claim 8, with the extract container positioned in selective fluid communication with each of the canisters for selectively collecting the extract solution from each of the canisters.

13. The system of claim 1, further comprising a solvent collection container in fluid communication with the extract container and the solvent source, the solvent collection container positioned to receive the recycled solvent and direct the recycled solvent towards the solvent source.

14. The system of claim 1, further comprising a heating element thermally coupled with the extract container, the heating element operable for heating the extract container to a distilling temperature, the distilling temperature being at least a boiling point of the solvent, but below a boiling point of the essential oil, thereby separating, in the extract container, the extract solution into a post-extraction portion of the essential oil and the recycled solvent.

15. The system of claim 1, with the coolant comprising a refrigerant.

16. The system of claim 1, further comprising a coolant pump positioned to direct the coolant through the coolant line.

17. The system of claim 1 further comprising a coolant fluid circuit that is partially defined by the coolant line.

18. The system of claim 1, with the canister adapted to contain at least a lavender plant material.

19. A system for extracting a solute from a source material, the system comprising: a solvent source container positioned to provide solvent that includes at least a hydrocarbon solvent; canisters that each contain a plant material, each canister in selective fluid communication with the solvent source container, thereby producing an extract solution of the solvent and the solute; an extract container positioned to be in selective fluid communication with each of the canisters for receiving and collecting at least a portion of the extract solution as a collected extract solution; a heating element thermally coupled with the extract container for heating the extract container to at least a distilling temperature, the distilling temperature being at least a boiling point of the solvent, but below a boiling point of the solute, thereby separating, in the extract container, the collected extract solution into at least a post-extraction portion of the solute and a post-extraction portion of the solvent; a solvent collection line positioned for directing, in a first direction along the solvent collection line, the post-extraction portion of the solvent from the extract container and towards the solvent source container; a coolant line for directing coolant that is thermally coupled to at least a portion of the solvent collection line for cooling the post-extraction portion of the solvent that is flowing through the solvent collection line in the first direction, with at least one of the solvent collection line and the coolant line defining a coiled portion along the portion of the solvent collection line for cooling the post-extraction portion of the solvent; and at least one pump positioned for at least directing the post-extraction portion of the solvent, the at least one pump

further positioned downstream of the extract container and upstream of the portion of the solvent collection line for cooling the post-extraction portion of the solvent.
